

Tracial Joint Spectral Measures for Compact Self-Adjoint Operators

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2026-07-03

Source And Target

The source is Otte Heinavaara, *Tracial joint spectral measures*, arXiv:2310.03227. For Hermitian matrices A, B , the source constructs a positive measure $\mu_{A,B}$ on $\mathbb{R}^2$ satisfying

$$\mathrm{tr} H(f)(xA + yB) = \int_{\mathbb{R}^2} f(ax + by) d\mu_{A,B}(a, b), \quad (1)$$

where

$$H(f)(s) = \int_0^1 \frac{1-t}{t} f(ts) dt.$$

The same paper notes that $\mu_{A,B}$ is necessarily unique away from the origin. Section 1.8 lists as future work the “existence and uniqueness of the measures $\mu_{A,B}$ for compact self-adjoint operators A and B .”

1.8. Future work. Our structural result opens interesting research directions. Some of these will be pursued in our forthcoming work [\[Hei23\]](#), which in particular will study the following topics:

- Geometric and regularity properties of μ_c , μ_s and their supports.
- Further consequences and reformulations of the main identity [\[3\]](#).
- The structure of measures $\mu_{A,B}$ for small matrices A and B .
- Existence and uniqueness of the measures $\mu_{A,B}$ for compact self-adjoint operators A and B .
- Optimality results which show various ways in which our main result cannot be improved.
- Establishing a relationship to hyperbolic polynomials in the sense of Gårding [\[Gär59\]](#).

Idea Of The Proof

Compress A and B to increasing finite-dimensional subspaces. The source gives a positive tracial joint spectral measure for each compression. Although these measures need not have a uniformly bounded total mass near the origin, compactness gives exactly the bound that is needed on every annulus away from the origin: if a compact set $K \subset \mathbb{R}^2 \setminus \{0\}$ is covered by finitely many strips where $|xa + yb|$ is bounded below, then the source identity bounds $\mu_n(K)$ by finitely many traces of functions of compact operators that vanish near zero. Those traces are uniformly bounded because only finitely many eigenvalues of a compact operator can stay above a fixed threshold. Vague compactness therefore gives a Radon limit on the punctured plane. The trace identity passes to the limit by norm convergence of compressions and finite-rank spectral stability. Finally, the projection identities determine the measure: ridge functions $f(xa + yb)$ with $f \in C_c(\mathbb{R} \setminus \{0\})$ linearly approximate every compactly supported continuous function on the punctured plane.

Theorem 1 (Compact self-adjoint tracial joint spectral measures). *Let A and B be compact self-adjoint operators on a separable Hilbert space. Then there exists a unique positive Radon measure*

$$\mu_{A,B} \quad \text{on} \quad \mathbb{R}^2 \setminus \{0\}$$

with support contained in

$$\{(a, b) \in \mathbb{R}^2 : |a| + |b| \leq \|A\| + \|B\|\}$$

such that for every $x, y \in \mathbb{R}$ and every $f \in C_c(\mathbb{R} \setminus \{0\})$,

$$\operatorname{tr} H(f)(xA + yB) = \int_{\mathbb{R}^2 \setminus \{0\}} f(ax + by) d\mu_{A,B}(a, b), \quad (2)$$

where

$$H(f)(s) = \int_0^1 \frac{1-t}{t} f(ts) dt.$$

The same identity holds for bounded Borel f with compact support contained in $\mathbb{R} \setminus \{0\}$.

Lemma 1 (Trace continuity away from zero). *Let T_n, T be compact self-adjoint operators with $T_n \rightarrow T$ in operator norm. If $h : \mathbb{R} \rightarrow \mathbb{C}$ is continuous on a neighborhood of $[-M, M]$, where $M > \sup_n \|T_n\| + \|T\|$, and h vanishes on a neighborhood of 0, then $h(T_n)$ and $h(T)$ are finite-rank for all large n and*

$$\operatorname{tr} h(T_n) \rightarrow \operatorname{tr} h(T).$$

Proof. Choose $\eta > 0$ such that $h = 0$ on $[-\eta, \eta]$. Since T is compact, the spectral subspace of T for $\mathbb{R} \setminus (-\eta/2, \eta/2)$ has finite dimension, say N . For all large n , norm spectral stability or the min-max principle gives

$$\operatorname{rank} 1_{\mathbb{R} \setminus (-\eta, \eta)}(T_n) \leq N.$$

The continuous functional calculus gives $\|h(T_n) - h(T)\| \rightarrow 0$. The difference has rank at most $2N$, hence

$$|\operatorname{tr} h(T_n) - \operatorname{tr} h(T)| \leq 2N \|h(T_n) - h(T)\| \rightarrow 0.$$

□

Lemma 2 (Ridge uniqueness on the punctured plane). *Let μ and ν be positive Radon measures on $\mathbb{R}^2 \setminus \{0\}$ with support contained in a common bounded set. Suppose that for every $u = (x, y) \in \mathbb{R}^2$ and every $f \in C_c(\mathbb{R} \setminus \{0\})$,*

$$\int f(u \cdot z) d\mu(z) = \int f(u \cdot z) d\nu(z).$$

Then $\mu = \nu$.

Proof. It suffices to show that the linear span of the ridge functions $z \mapsto f(u \cdot z)$, with $f \in C_c(\mathbb{R} \setminus \{0\})$, is dense on every closed ball in the class of continuous functions that vanish in a neighborhood of the origin.

Let $\varphi \in C_c(\mathbb{R}^2 \setminus \{0\})$, and extend it continuously to the origin by setting $\varphi(0) = 0$. Fix a ball $\{|z| \leq R\}$ containing the common supports of the measures. Since φ vanishes on a neighborhood of 0, for any $m \geq 1$ the function

$$\psi(z) = \begin{cases} \varphi(z)/|z|^{2m}, & z \neq 0, \\ 0, & z = 0, \end{cases}$$

is continuous on this ball. By Weierstrass, choose a polynomial Q such that $P(z) = |z|^{2m}Q(z)$ approximates φ uniformly on the whole ball. The polynomial P has no homogeneous terms of degree 0. Each homogeneous polynomial of positive degree is a finite linear combination of powers $(u \cdot z)^d$, because symmetric tensors are spanned by pure powers. Thus P is a finite sum of such ridge polynomials, all of positive degree.

For each term $(u \cdot z)^d$, replace t^d on the bounded interval $[-\|u\|R, \|u\|R]$ by a function $f_{u,d} \in C_c(\mathbb{R} \setminus \{0\})$ that agrees with t^d except possibly for $|t| < \tau$, and is zero near 0. Taking τ small makes the replacement uniformly small on the whole ball because $d \geq 1$. The resulting finite sum of allowed ridge functions approximates φ uniformly on the common bounded support and also vanishes near the origin. Equality of integrals for ridge functions therefore implies equality of integrals for φ . $\square$

Proof of Theorem 1. Let P_n be an increasing sequence of finite-rank orthogonal projections that converges strongly to the identity. Since A and B are compact,

$$A_n = P_n A P_n \rightarrow A, \quad B_n = P_n B P_n \rightarrow B$$

in operator norm. Let $\mu_n = \mu_{A_n, B_n}$ be the finite-dimensional measure from the source, viewed as a measure on $\mathbb{R}^2 \setminus \{0\}$.

We first prove uniform local boundedness. Let $K \subset \mathbb{R}^2 \setminus \{0\}$ be compact. There are finitely many directions $u_\ell = (x_\ell, y_\ell)$ and a number $\eta > 0$ such that every $z \in K$ satisfies $|u_\ell \cdot z| \geq 2\eta$ for at least one ℓ . Choose $f_\ell \in C_c(\mathbb{R} \setminus \{0\})$, $0 \leq f_\ell \leq 1$, such that $f_\ell(t) = 1$ on the possible values of $u_\ell \cdot z$ with $z \in K$ and $|u_\ell \cdot z| \geq 2\eta$. The source identity gives

$$\mu_n(K) \leq \sum_\ell \int f_\ell(u_\ell \cdot z) d\mu_n(z) = \sum_\ell \text{tr } H(f_\ell)(x_\ell A_n + y_\ell B_n).$$

For each ℓ , the operators $x_\ell A_n + y_\ell B_n$ converge in norm to the compact operator $x_\ell A + y_\ell B$, and $H(f_\ell)$ is continuous and vanishes near 0. Lemma 1 gives a uniform bound on these traces. Hence $\sup_n \mu_n(K) < \infty$.

By the usual diagonal compactness argument for Radon measures on the locally compact second countable space $\mathbb{R}^2 \setminus \{0\}$, a subsequence of (μ_n) converges vaguely to a positive Radon measure μ . The finite dimensional support bound in the source gives

$$\text{supp } \mu \subseteq \{(a, b) : |a| + |b| \leq \|A\| + \|B\|\}.$$

Fix $x, y \in \mathbb{R}$ and $f \in C_c(\mathbb{R} \setminus \{0\})$. If $(x, y) \neq (0, 0)$, then $z \mapsto f(xa + yb)$ has compact support in $\mathbb{R}^2 \setminus \{0\}$ after intersecting with the above bounded support. Vague convergence and the finite-dimensional identity yield

$$\int f(xa + yb) d\mu = \lim_{n \rightarrow \infty} \int f(xa + yb) d\mu_n = \lim_{n \rightarrow \infty} \text{tr } H(f)(xA_n + yB_n).$$

The last limit equals $\text{tr } H(f)(xA + yB)$ by Lemma 1, because $H(f)$ vanishes near 0. The case $(x, y) = (0, 0)$ is immediate. Thus μ satisfies (2).

If ν is another positive Radon measure satisfying the same identity, then the source support argument applies verbatim: testing with non-negative functions supported outside $[-M, M]$ shows that ν is supported in $|a| + |b| \leq \|A\| + \|B\|$. The identity gives equality of all ridge integrals for μ and ν , so Lemma 2 gives $\mu = \nu$. Consequently the vague limit is independent of the chosen subsequence and of the approximating projections.

Finally, bounded Borel tests supported away from 0 follow from the continuous case by the monotone class theorem, applied to the finite projection measures and to the finite spectral measures of $xA + yB$ on the part of the spectrum where $H(f)$ can be nonzero. $\square$

Corollary 1 (Schatten moment identity). *Let A and B be compact self-adjoint operators and let $p > 0$. If $xA + yB \in S_p$, then*

$$\mathrm{tr} |xA + yB|^p = p(p+1) \int_{\mathbb{R}^2 \setminus \{0\}} |ax + by|^p d\mu_{A,B}(a, b).$$

In particular this holds for all x, y whenever $A, B \in S_p$.

Proof. Apply Theorem 1 to the increasing cutoffs $f_m(t) = |t|^p \chi_m(t)$, where χ_m is supported away from zero and $\chi_m(t) \uparrow 1$ for $t \neq 0$. Since $H(|\cdot|^p)(s) = |s|^p/(p(p+1))$, monotone convergence on both the spectral side and the measure side gives the identity. $\square$

Verification Notes

The proof is analytic rather than computational. The script `code/verify_diagonal_non_hs.py` checks a commuting diagonal compact model with eigenvalues comparable to $j^{-1/2}$, hence not Hilbert-Schmidt. It prints finite annular masses for the explicit segment-measure formula while the partial second moments grow, illustrating why the full compact theorem must be formulated on the punctured plane with local annular finiteness.

Novelty Check And Limitations

Cheap run indexes were searched for arXiv:2310.03227, tracial joint spectral measure keywords, compact self-adjoint operators, and $\mu_{A,B}$. A bounded web search was also run for “Properties of Tracial Joint Spectral Measures,” “Tracial joint spectral measures compact self-adjoint,” and close variants. No available later answer was found; the source bibliography lists the follow-up as in preparation.

The result proves uniqueness on $\mathbb{R}^2 \setminus \{0\}$, matching the source’s finite-dimensional uniqueness-away-from-zero convention. It does not assign canonical mass at the origin, where the finite-dimensional construction is already not determined by the identities.

References

References

- [1] O. Heinavaara, *Tracial joint spectral measures*, arXiv:2310.03227, 2023.